Homework 10: Privacy and Security (15 points)

Due to Gradescope by 11:45 PM on Tuesday, November 15th
You need to submit a pdf to Gradescope; failure to assign questions to pages will result in a 10% deduction on your grade

Homework Goal: Practice various types of encryption and decryption; understand the importance of security and different algorithms

Step	Simple Password Hash	Example: result using "c1ph3r" as input
1	Replace each letter with its number (a → 1, etc.), keep digits as is	3 1 16 8 3 18 c1ph3r
2	Add up all the *digits* in the result of step 1	3+1+1+6+8+3+1+8 = 31
3	Reverse the digits in the result of step 2	13
4	Find the remainder of: step 3 result / 20	13%20 = 13
5	Multiply the result of step 4 by the number of digits in the result of step 2	13*2 = 26
6	Exchange each digit with its character in the result of step 5 (replace 0's with Z's)	bf

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

1. Using the simple hash algorithm above, find the encrypted forms of the following passwords. (3 points)

(a) abc1234 → 1231234 → 1+2+3+1+2+3+4 → 16

61%20 = 1 • 2 = 2 → B

(b) password → 16+1+19+19+23+15+18+4 → 115

511%20 = 11 • 3 = 33 → CC

(c) ilove101 → 9+12+15+22+5+1+10+1 → 65

56%20 = 16 • 2 = 32 → CB

2. The password for a Zoom account is 4 characters long, consisting of letters (A-Z, a-z), digits (0-9), and these four symbols: ! @ # $ The characters can occur in any order and repeats are allowed. (2 points)
   (a) How many different passwords are possible?

$26+26+10+4=66$

$4^{66} = 5.4 \times 10^{39}$

   (b) If you write a program that can test 4,000 passwords per second, about how many **hours** would it take you to try every possible password?

$4{,}000 \text{ per/s} \cdot 60 = 240{,}000 \text{ per/min} \cdot 60 = 14400000 \text{ per/hour}$

$4^{66} / 14400000 = 3.8 \times 10^{32}$ Hours

3. Consider the encryption algorithms we've talked about in class. (1 point)
   (a) Which algorithm(s) are symmetric encryption algorithm(s)?

DES, AES

   (b) Which algorithm(s) are asymmetric encryption algorithm(s)?

RSA

4. Given the following sample email, what are two clues that indicate this might be a phishing email? (1 point)

Dear customer: ← Not using name

We write to inform you that you account will soon expire. In order to keep it up to date, please go this link and enter your password:

http://zoorn.us Not Zoom, spelt wrong

Sincerly,

Your Zoom Team

5. Your friend tells you that a Caesar Cipher is more secure when you use larger numbers for the key (like 500 or 50,000). Do you believe them? Why or why not? (1 point)

No I do not agree because there is only 26 different ways to encrypt a letter and that is rather limited.

6. My cat, Bones, encoded a message for me using a Meow-sar Cipher (Caesar Cipher for cats): "IHHG PH WXQD". Unfortunately, she is unable to tell me the shift (how I even got the message from her, I'll never know). Determine the shift and decode the message from Bones (by hand, not through an online site). (1 point)

IHHG PH WXQD Shift of 3

ABCDEFGHIJKLMNOPQRSTUVWXYZ
XYZABCDEFGHIJKLMNOPQRSTUVW

FEED ME TUNA

7. What is a Trojan? Explain an example of a real Trojan. (1 point)

A program that seems to do a legitamate task while also breaking security. An example is where a user downloads an app and runs that app. the app then creates a backdoor.

8. Your dad has taken to sending you block-encoded messages to help you prepare for your upcoming CS exam. Today's dad joke (almost worse than your instructor's CS jokes) has arrived in your inbox, and it reads "I can sum up 2021 in one word… JKVJ" To determine his punchline, decrypt the message "JKVJ". Show your work. (3 points)

$$X = \begin{bmatrix} 3 & 2 \\ 7 & 5 \end{bmatrix} \qquad X' = \begin{bmatrix} 5 & 24 \\ 19 & 3 \end{bmatrix}$$

2x2

$$S = \begin{bmatrix} 10 & 11 \\ 22 & 10 \end{bmatrix}$$ 2x2

$$(X' \cdot S) \% 26 =$$

$$\begin{bmatrix} 578 & 295 \\ 256 & 234 \end{bmatrix} \% 26 = \begin{bmatrix} 6 & 9 \\ 22 & 5 \end{bmatrix}$$

FIVE

9. Bob and Alice both have their own public key and private key. Alice wants to send Bob a message that he will know was sent from her. With what key should Alice encrypt the message? Would this way of encryption prevent a third party from being able to read Alice's message? (2 points)

Alice should sign with her private key and then Bob will authenticate the message with Alice Public key. No because Alice isnt encrypting the message with Bob's pulblic key. Anyone with Alice's public key can authenticate the message.